



The Linear Probability Model (LPM) Transcript

Hi everyone, let's learn in this video about the linear probability models and those types of models which are specifically used in the cases where our output variable is categorical. You have learned about cases where output variable is continuous, you learned about linear regression, multiple linear regression. And what if you have the output as yes or no, or true or false, or high, medium, low.

Those are categories. Those are not values or real numbers in terms of continuous values, right? So let's say 0.8, 0.6, 0.2, or 10,000, 100. They are not numerical.

They are not numeric values. They are categorical values. So in the cases where your output that you are trying to look for or that you are trying to predict is a categorical value, just like yes and no.

So let me give you an example. Let's say you have a data of a lot of people who have or have not cancer. Now you have evaluated who has cancer, who does not have cancer.

Now given that this data is here, how would you use this data to create a model such that any new individual who comes in, you should be able to sufficiently answer that the individual is not having cancer or the individual is very likely to have cancer. So because the decision is to understand whether the individual is going to have cancer or not, yes or no, hence we use the kind of models which are called as logit-based models or logistic regression models. That is the entire intention of this entire 3 or 4 set of videos we are going to learn in this entire exercise.

So I hope this entire video would give you a good understanding of how do we use logistic regression models for predicting binary outcomes or when the outcomes are actually not binary but there could be other outcomes as well. But we will start with a simple case when the outcomes are binary. Okay, yes, no, true, false, high, low and think of anything which has these binary outcomes.

So just to give you a start point, let's first discuss about what could be various different types of those scenarios when the output could be binary. Let's say will the customer purchase the product or not, whether someone who is currently active on a subscription plan, whether they will renew the subscription or not, whether the loan would be paid by the people who have taken it or not, whether the patient would respond to the treatment or not. Think of many such situations and if you have data to capture a model like these situations, then what is the method you should apply or you should innovate so that you get to the output probabilities or results that this is likely to happen, this is what exactly is likely to happen.

So that's the entire idea. So we model probability of getting a yes, let's say y is equals to 1 is yes or success. Okay, given all the input parameters, let's say for example, you have gone out for a test of flu and what is the probability of you having flu given you have certain symptoms, right, given your symptoms.



So exactly that is what we are going to look for. Now applying ordinary least square, so let's say linear regression models to the situation like this when y is categorical, what would happen in those cases, when you have situation like this, you actually can fit in a linear regression model in these situations, but what would happen if you think of you would have situations wherein your output estimate would be greater than 1, less than 0, but you are only looking for two values 0 or 1. You are only looking for situations wherein you can either say it is going to fail or pass true or not true, right, yes or no. So you are looking for predictions which are contained within 0 and 1 and not beyond 0 or lesser than 1, sorry, beyond 1 and lesser than 0. So we want to be in that situation wherein our predictions are actually saying either 0 or 1 and not something greater than that.

Something in between is still fine because we can relate to if a value is 0.8, we can say that there are high chances of being 1, the value is closer to 0.2, we are saying that there are more chances of being 0 or no, right, so or failure. So the problem with using linear regression models in those situations is the output would be non-sensical predictions because they will be non-bounded between 0 and 1. They will be outside, they can be outside the limits. Second, the assumption gets violated.

The errors are not normally distributed. If you remember one of the most important assumptions that we have in linear regression is to follow or the error should follow normally distributed with or they should be normally distributed with mean 0 and variance, constant variance σ^2 , right. Now given that the assumptions will not be followed here given that y itself is non-continuous.

Obviously it is not going to follow that assumption, okay. So these are some of the problems. That is why we do not use linear regression models in these situations when the output is binary.

So let us now understand in the next video what are the kind of things we can use till this point you know what exactly are we talking about, what exactly is the problem. Problem is to predict something which has binary output, yes, no, true, false, high, medium, not what not, customer going to subscribe or not, they are going to repay the loan or not, someone is going to have the disease after some point in time, yes or no, right. So you know the problem domain.

Now we are going to get into the details of how do we solve that problem.

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